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Unit – 5 : Session – 3 : SLO – 1 | Implementing Automata with the Python automata Library

■ I. Questions & Answers

1. Library — the automata-lib library (pip install automata-lib), specifically automata.fa.dfa.DFA and automata.fa.nfa.NFA, is used to define DFAs/NFAs in Python.
2. Acceptance check — `dfa.accepts_input("0101")` returns True/False; `dfa.read_input("0101")` returns the final state.
3. Defining a transition — transitions are given as a nested dictionary: `transitions={"q0": {"0": "q1", "1": "q0"}, "q1": {"0": "q1", "1": "q0"}}` — i.e. $\delta(\text{state}, \text{symbol}) = \text{next state}$.

■ II. Example

- `from automata.fa.dfa import DFA`
- `dfa = DFA(states={"q0", "q1"}, input_symbols={"0", "1"},`
- `transitions={"q0": {"0": "q1", "1": "q0"}, "q1": {"0": "q1", "1": "q0"}},`
- `initial_state="q0", final_states={"q1"})`
- `print(dfa.accepts_input("1010")) # True` — string ends with 0